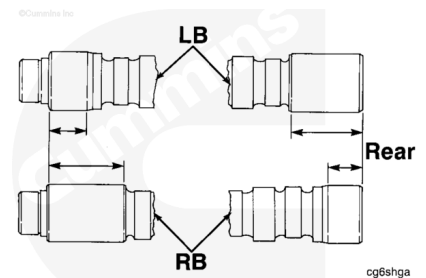


Trainings ▾

Install

⚠ CAUTION ⚠

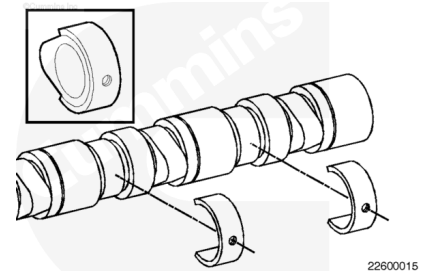
The right and left camshafts are different. They must be installed in the correct bank. If they are not installed correctly, the engine will be seriously damaged.



Note : The camshafts for the QSK45 and QSK60 are different. The installation procedure, however, is the same.

Make sure the correct camshaft is installed. Refer to the CPL for the engine being serviced. If the wrong camshaft is installed, the engine will **not** perform to specifications.

Use camshaft tool kit, Part Number 3163195. Install the camshaft installation guides **only** on the inner base circle of the valve lobes as defined and center the guides on the camshaft. Install all of the guides before removing or installing the camshaft.



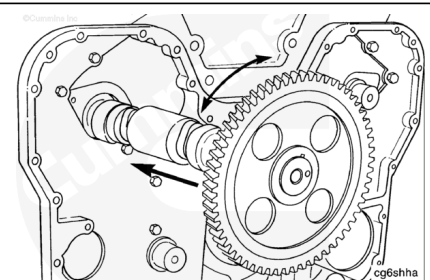
Install the camshaft installation guides, Part Number 3163195, on the valve lobes of the camshaft.

Install the camshaft pilot onto the rear of the camshaft.

Use clean engine oil to lubricate the camshaft and camshaft bushings.

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.



⚠ CAUTION ⚠

Do not damage the camshaft bushings when installing the camshaft. The bushing will fail.

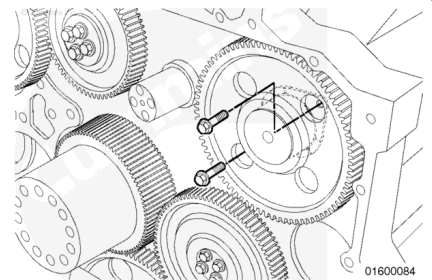
Install the camshaft by turning it slightly **clockwise** and **counterclockwise** as it is pushed into the block. Keep the guides turned downward to support the camshaft.

Align the thrust bearing, and install the two capscrews.

The narrow side of the thrust bearing plate is installed towards the outside of the engine block.

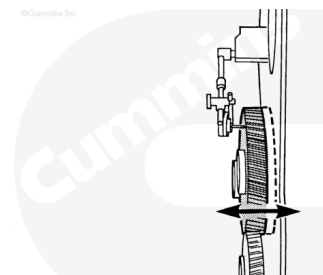
Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



Use a dial indicator, Part Number 3376050, and magnetic base, Part Number 3377399, to measure the camshaft's end clearance.

Camshaft End Clearance		
mm		in
0.15	MIN	0.006
0.33	MAX	0.013



If the clearance is **not** within specifications, check for foreign material or a piece of gasket between the thrust plate and the block.